

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 529 — Past Questions & Materials
by Renance

Course: EMT 529
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 500

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Paper 1 of 1 — EMT 529 SEPARATION TECHNIQUES IN ENVIRONMENTAL 2020,2021

Level 500 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY

SECOND SEMESTER EXAMINATION FOR 2020/2021 ACADEMIC SESSION

COURSE TITTLE: SEPARATION TECHNIQUES IN ENVIRONMENTAL ANALYSIS

COURSE CODE: EMT 529 TIME: 2 Hours 30minutes

INSTRUCTIONS: Answer four questions with at least one from each section

SECTION A

1. a) Explain the term separation techniques (7 4 marks)
b) In tabular form differentiate between mixture and compound (1@marks)
2. a) In a tabular form, state the basis of separation of mixtures and the corresponding technique or method used (10marks)
b) Explain two of the separation methods with one or more example(s). (7 4 marks)
3. a) Assuming you were to run an analysis and the only water available for use in that instance is saline water. Kindly explain how you can get a clean water from the available one using any separation technique. (7 % marks)
(b) Mention and explain the separation techniques that could be used in the process of wastewater treatment. (10marks)

SECTION B

4. a) Define the term Solvent extraction
b) Using different formula expressions. Discuss the terms Partition coefficient (IX) and Distribution Ratios (D) with respect to Solvent extraction
c) Name seven Extraction techniques that you are familiar with and state the principle behind each one.
- ee a) Chromatography depends on the combination of the two phases i.e, the stationary phase and the mobile phase. Using your chromatography knowledge fill in the empty space below: